

2026.2

Algo. & Data Structure II

DCA3702

Prof. Ivanovitch Silva

UFRN DCA

Prazo Para Conclusão (em semestres): Mínimo: 5 **Médio:** 10 **Máximo:** 14

Optativas | Complementares | 1º Nível | 2º Nível | 3º Nível | 4º Nível | 5º Nível | 6º Nível | 7º Nível | 8º Nível | 9º Nível | 10º Nível

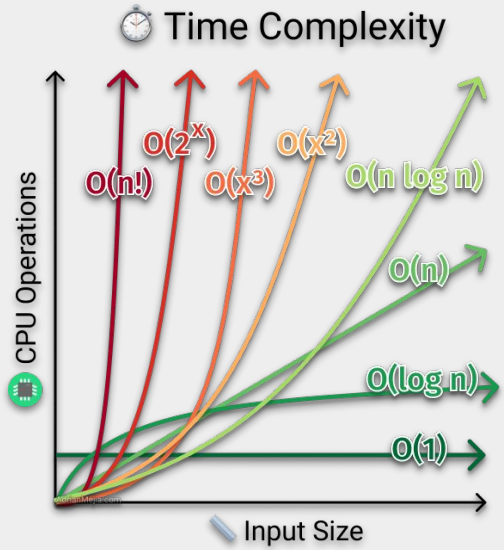
7º Nível

DCA3701 - PROJETO DE SISTEMAS DE CONTROLE - 90h	*Obrigatória*
DCA3702 - ALGORITMOS E ESTRUTURAS DE DADOS II - 60h	*Obrigatória*
DCA3703 - PROGRAMAÇÃO PARALELA - 45h	*Obrigatória*
DCA3704 - SISTEMAS DISTRIBUÍDOS - 45h	*Obrigatória*
DCA3705 - AUTÔMATOS E LINGUAGENS FORMAIS - 60h	*Obrigatória*
DCA3706 - SISTEMAS EMBARCADOS - 60h	*Obrigatória*

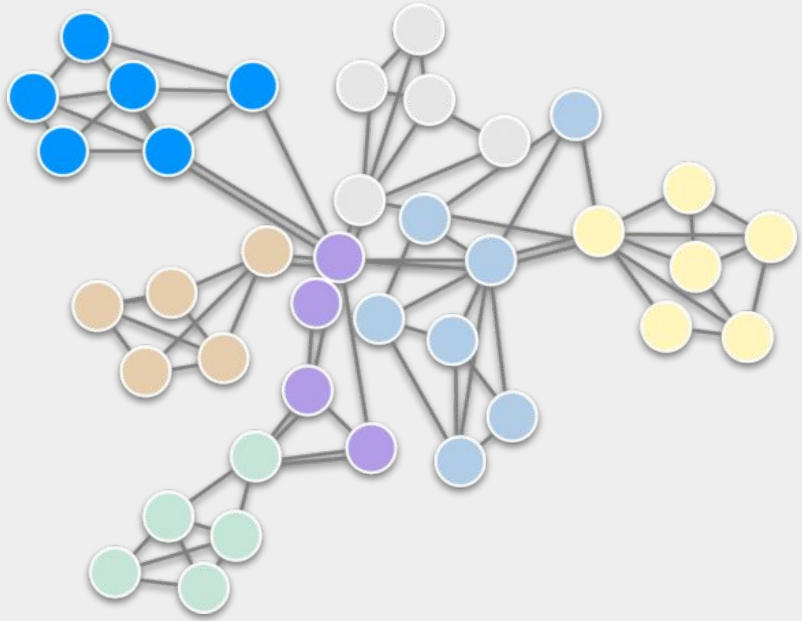
Carga Horária Total: 360h

Course Outline

To enable students to model engineering problems as graphs, implement and analyze fundamental algorithms, interpret structural properties of real-world networks, and develop contemporary solutions



Algorithm Complexity



Graph

Documentation

Search ⌘ K

- Computers
- **Accessories**
  - SD Cards
  - SSDs
  - SSD Kit
  - M.2 HAT+
  - Touch Display 2
  - Monitor
  - Camera
  - AI Camera
  - Keyboard and mouse
  - USB accessories
  - Build HAT
  - Sense HAT
  - AI Kit
  - **AI HATs**
    - How it works

AI HATs

AI HATs are add-on boards for Raspberry Pi 5 that come with a built-in AI accelerator chip: the Hailo neural processing unit (NPU). The Hailo NPU allows Raspberry Pi 5 to run hardware-accelerated AI models locally, removing the need to send data to a remote cloud server for processing. This edge AI approach improves performance, reduces latency, and helps to keep data private. For information about how AI HATs integrate with Raspberry Pi 5 to provide hardware-accelerated AI inference, see How it works.

AI HATs use the newer Raspberry Pi HAT specification, indicated with a plus sign (**+**). There are two AI HAT models: **AI HAT+** and **AI HAT+ 2**. For more information, see Models. These AI HATs are available in different performance variants, measured in tera-operations per second (TOPS) of inference performance. AI HAT+ is available in a **13 TOPS** or a **26 TOPS** variant; AI HAT+ 2 is available in a single **40 TOPS** variant. For more information, see Performance (TOPS) variants.



Figure 1.
The 26 tera-operations per second (TOPS) Raspberry Pi AI HAT+

On this page

- How it works
- Models
- Performance (TOPS) variants
- Physical specifications
- Hardware assembly
- Next steps: Run AI models

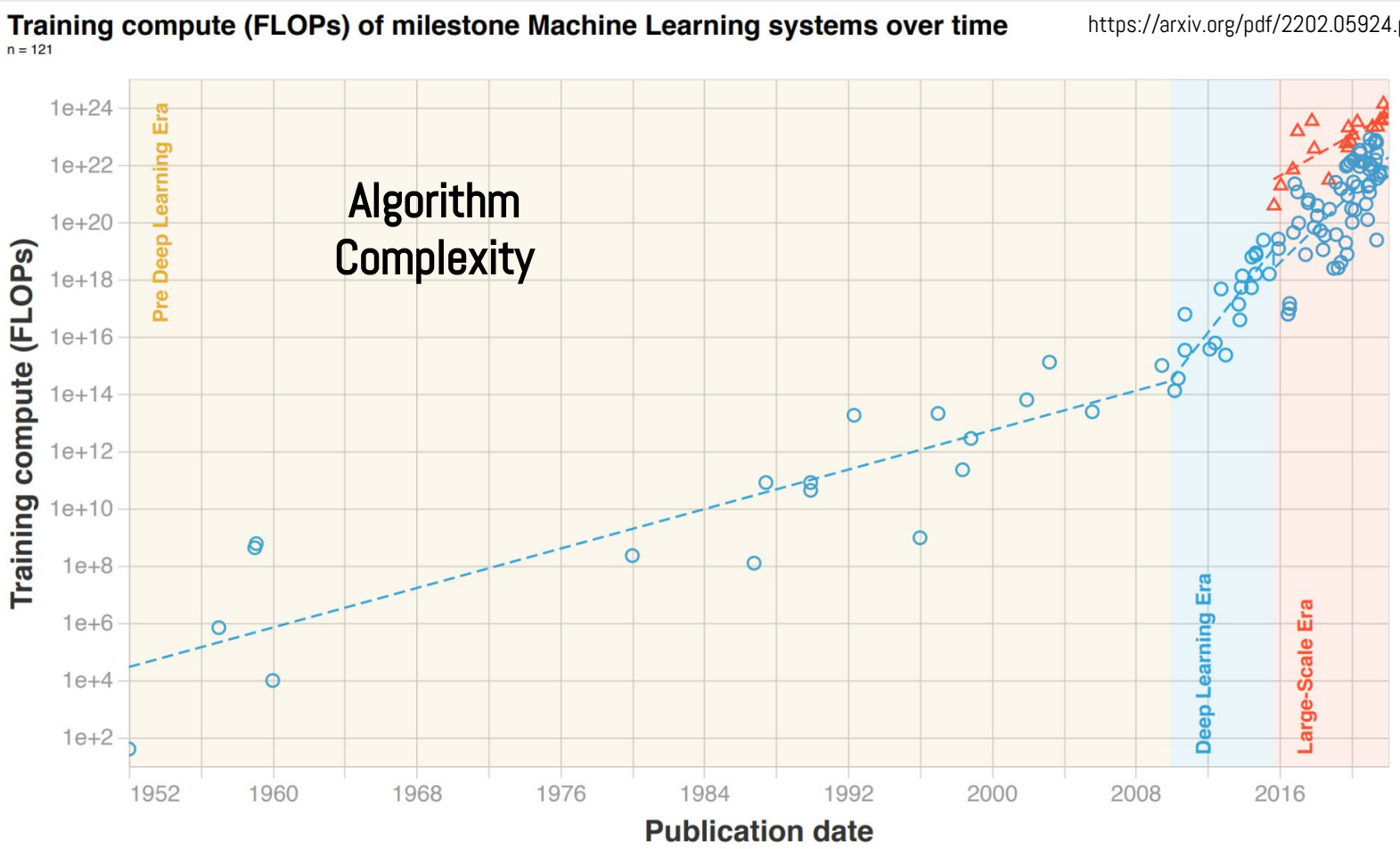
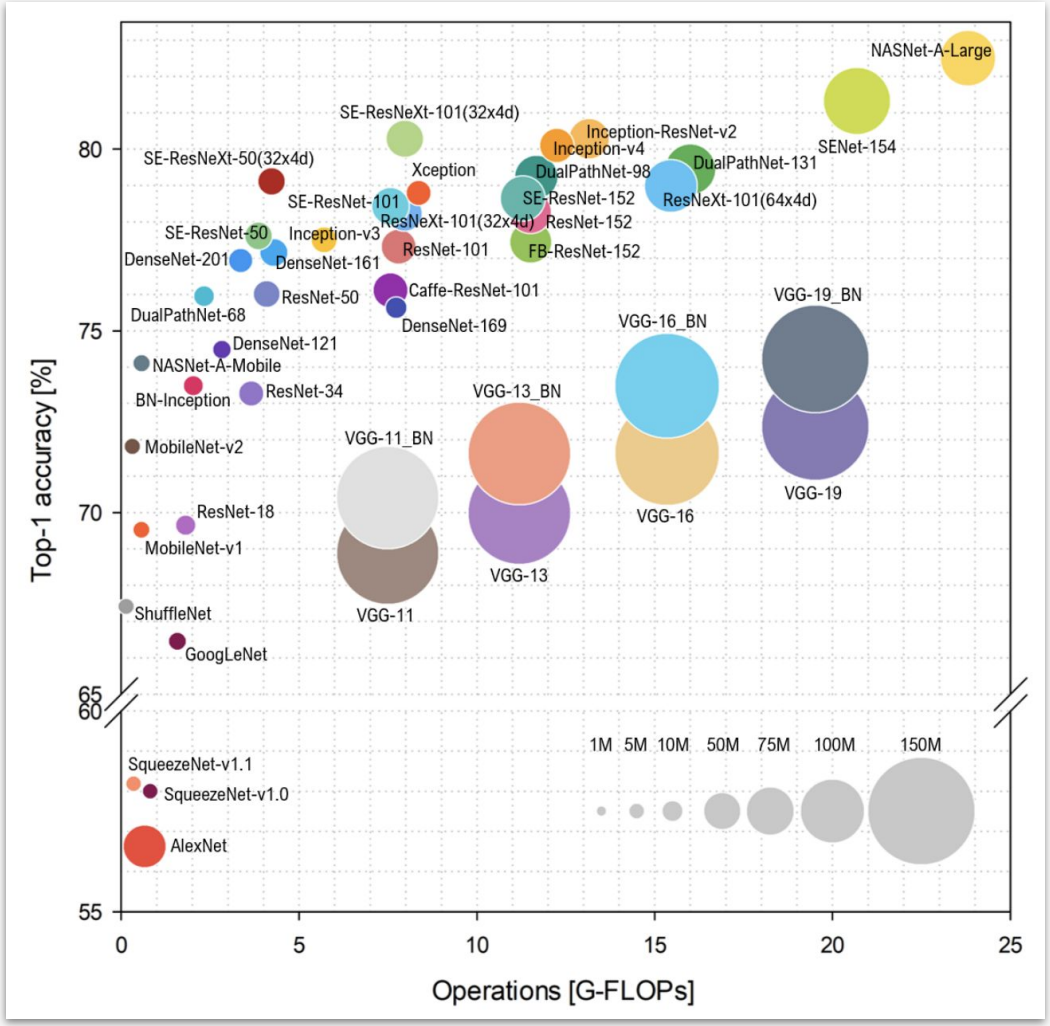


Figure 1: Trends in $n = 121$ milestone ML models between 1952 and 2022. We distinguish three eras. Notice the change of slope circa 2010, matching the advent of Deep Learning; and the emergence of a new large-scale trend in late 2015.

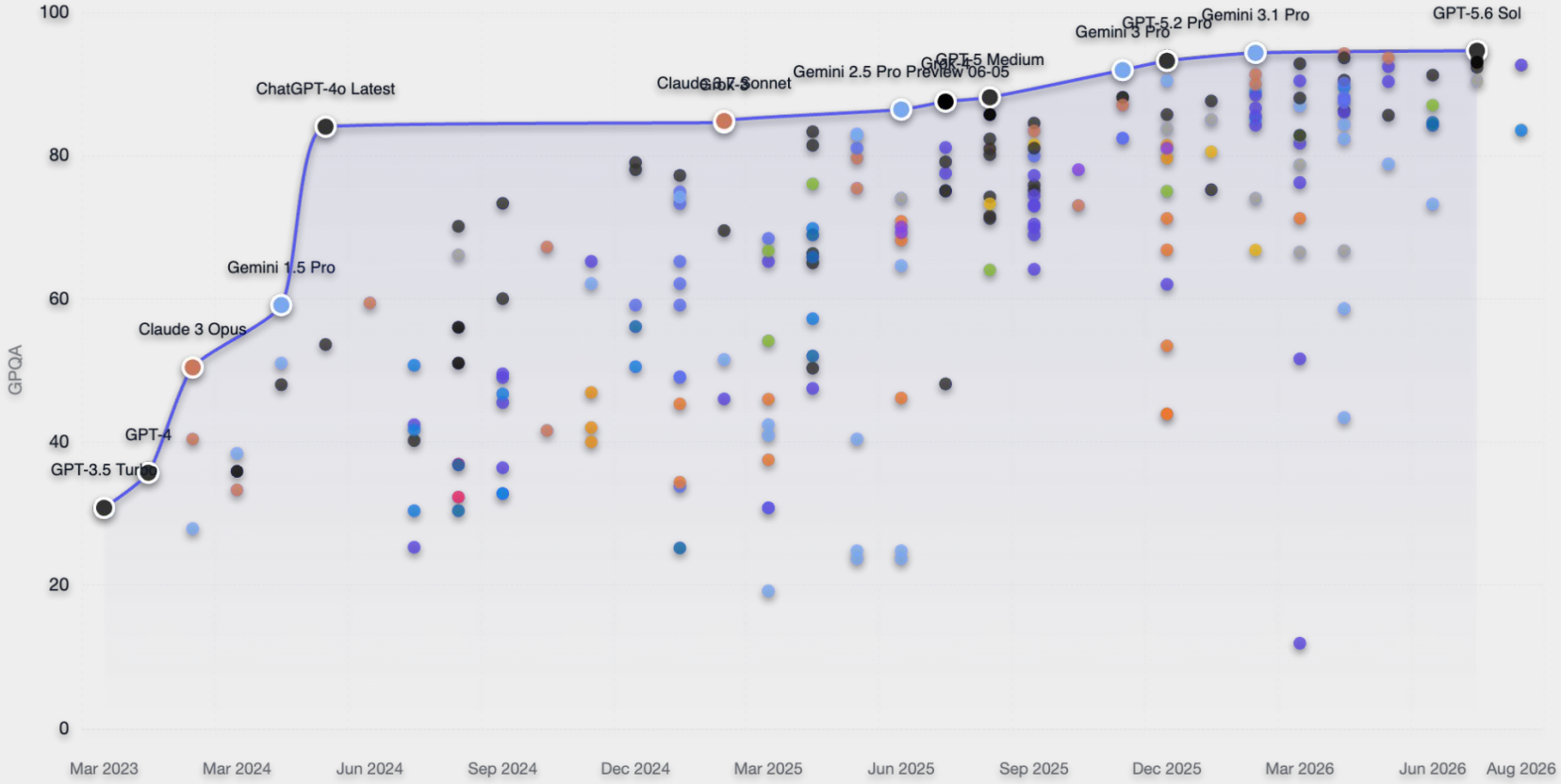


Algorithm Complexity

ML Model Evolution ImageNet

Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, "Benchmark analysis of representative deep neural network architectures". IEEE Access, vol. 6, 2018.

LLM Leaderboard



mlco2 / codecarbon Public

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Code Issues 108 Pull requests 27 Discussions Actions Projects Security and quality Insights

master 193 Branches 74 Tags Go to file Code

benoit-cty Merge pull request #1284 from mlco2/docs/linux_rapl_and_ans... 065d0e6 · 18 hours ago 2,891 Commits

.github	chore: use official pre-commit action (#1297)	3 days ago
act	Pin versions of actions by hash (#890)	last year
carbonserver	chore: upgrade potentially vulnerable libs (#1296)	4 days ago
codecarbon	feat: add Apple M2/M3/M4 chip TDP support (#1256)	3 days ago
deploy	Update RAPL access and Ansible deploy	18 hours ago
docker	cleaning	last year
docs	Update RAPL access and Ansible deploy	18 hours ago
examples	Fix multiple-counting	last week
requirements	chore: upgrade potentially vulnerable libs (#1296)	4 days ago
scripts	Refactor link checking script for improved functionality	4 months ago
tests	feat: add Apple M2/M3/M4 chip TDP support (#1256)	3 days ago
webapp	chore: upgrade potentially vulnerable libs (#1296)	4 days ago
.editorconfig	Import Github actions and contributions configuration fro...	6 years ago
.env.example	feat: New docker-compose setup	2 years ago
.flake8	Refactor to reduce complexity	9 months ago
.gitattributes	feat: add .json and .csv to .gittatributes	3 years ago

About

Track emissions from Compute and recommend ways to reduce their impact on the environment.

docs.codecarbon.io/

ai-ethics carbon-emissions carbon-footprint co2-emissions energy-consumption energy-efficiency fairness sustainability

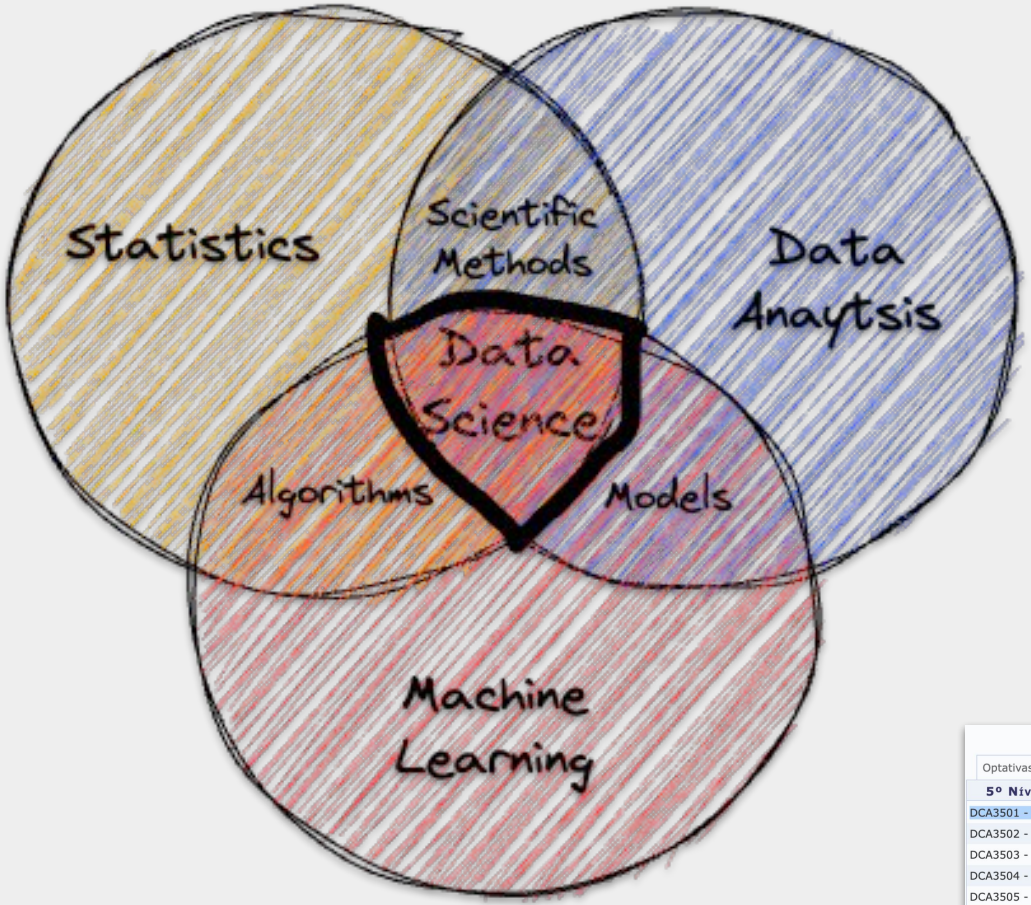
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And about graph?
Where and how use them?



Data Science is an interdisciplinary field that use scientific methods, processes, algorithms and systems to extract knowledge and insights from structure and unstructured data.

Data Scientists use data to answer questions.

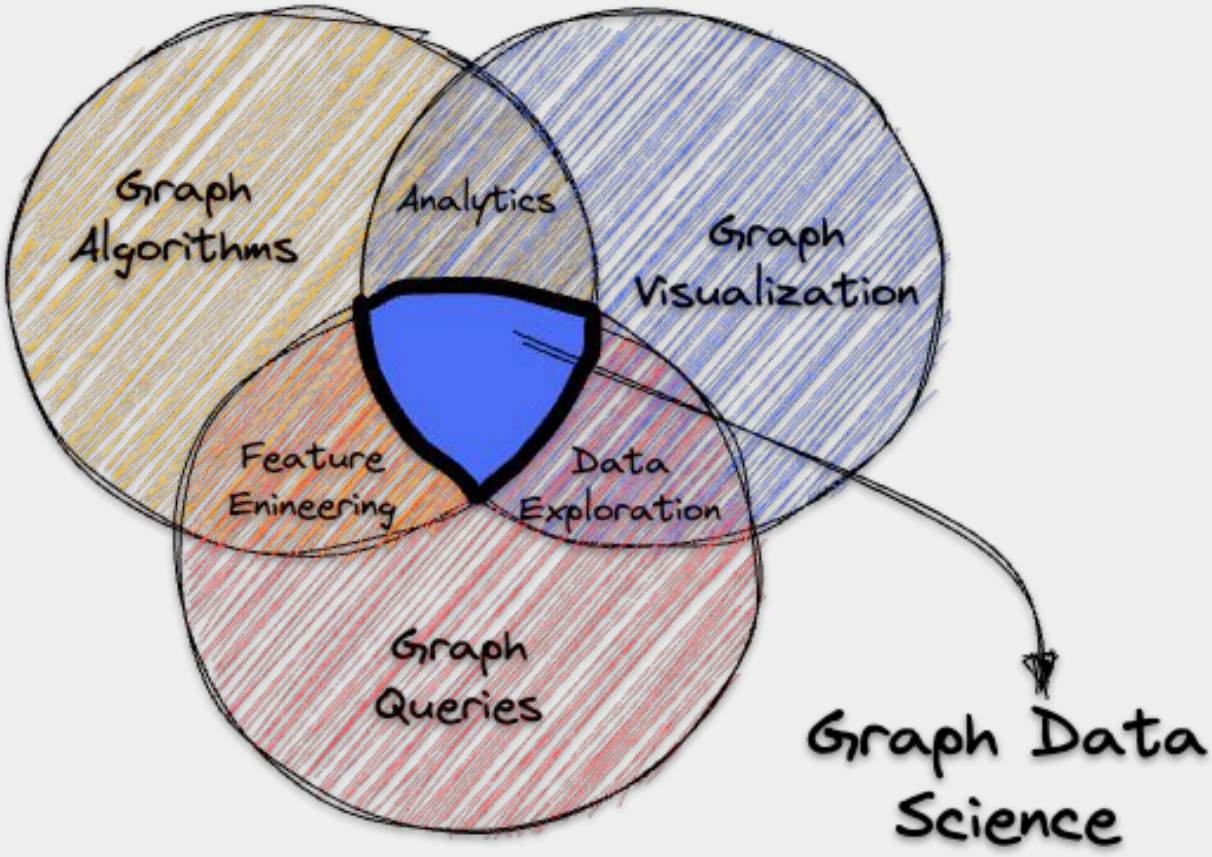
Prazo Para Conclusão (em semestres): Mínimo: 5 **Médio:** 10 **Máximo:** 14

Optativas | Complementares | 1º Nível | 2º Nível | 3º Nível | 4º Nível | 5º Nível | 6º Nível | 7º Nível | 8º Nível | 9º Nível | 10º Nível

5º Nível

DCA3501 - CIÊNCIA DE DADOS - 60h	*Obrigatória*
DCA3502 - SINAIS E SISTEMAS - 60h	*Obrigatória*
DCA3503 - ALGORITMOS E ESTRUTURAS DE DADOS I - 60h	*Obrigatória*
DCA3504 - OTIMIZAÇÃO DE SISTEMAS - 60h	*Obrigatória*
DCA3505 - SISTEMAS OPERACIONAIS - 60h	*Obrigatória*
DCA3506 - ELETRÔNICA - 75h	*Obrigatória*

Carga Horária Total: 375h



Graph Data Science is an science-driven approach to gain knowledge from the relationships and structures in data, typically to power predictions.

Data Scientists use relationships to answer questions.

Building and Environment 274 (2025) 112748

Contents lists available at ScienceDirect

Building and Environment

journal homepage: www.elsevier.com/locate/buildenv

What is a Digital Twin anyway? Deriving the definition for the built environment from over 15,000 scientific publications

Mahmoud Abdelrahman [a], Edgardo Macatulad [a,b], Binyu Lei [a], Matias Quintana [c], Clayton Miller [d], Filip Biljecki [a,e,*]

[a] Department of Architecture, National University of Singapore, 4 Architecture Drive, Singapore, 117566, Singapore
[b] Department of Geodetic Engineering, University of the Philippines, Diliman, Quezon City, 1101, Philippines
[c] Future Cities Lab Global Programme, Singapore-ETH Centre, CREATE campus, #06-01 CREATE Tower, Singapore, 138602, Singapore
[d] Department of the Built Environment, National University of Singapore, 4 Architecture Drive, Singapore, 117566, Singapore
[e] Department of Real Estate, National University of Singapore, 15 Kent Ridge Dr, Singapore, 119245, Singapore

ARTICLE INFO

Keywords:
Digital twin
Terminology
NLP
LLMs
Building digital twin
Urban digital twin
City digital twin

ABSTRACT

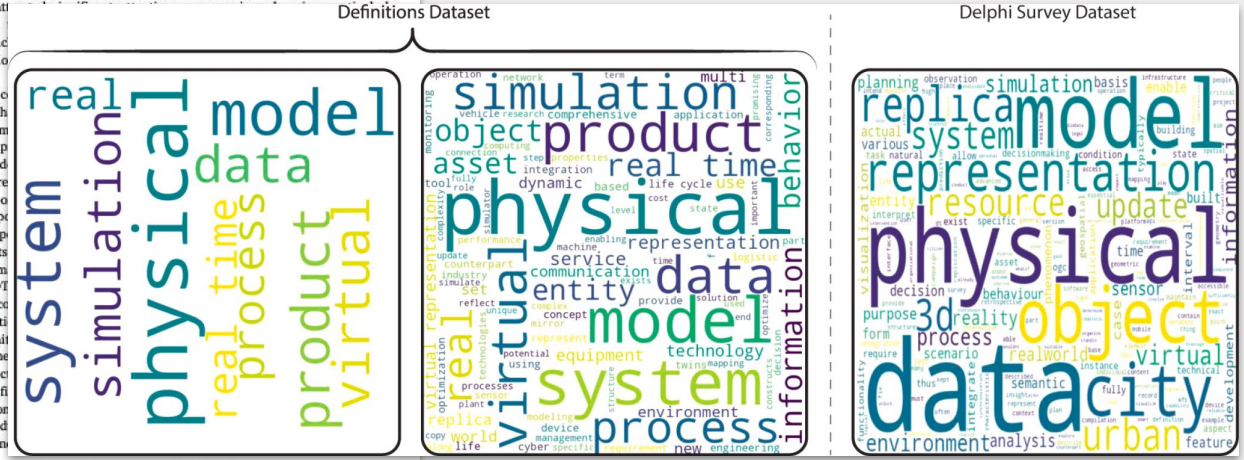
The concept of Digital Twins (DT) has att
within the built environment. However,
consensus remains out of reach. The lac
their conceptualization and implementatio
practitioners.

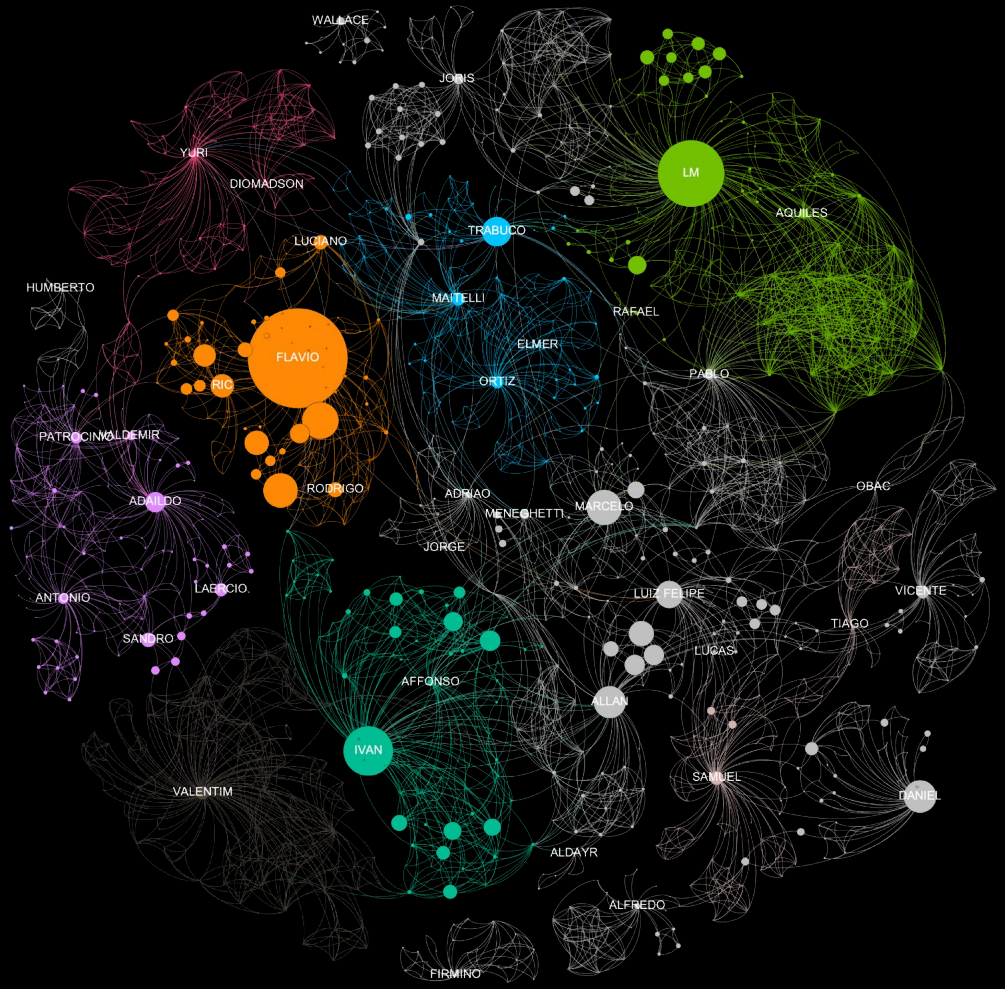
We employed Natural Language Proc
definitions of DTs from a corpus of more th
compares these findings with insights from
concurrence on the components that comp
domains, contrasting them with those that d
twin definitions over time and across differe
perspectives. We extracted the main compo
analysis. Subsequently, we identified compe
test to assess the significance of each comp

Our analysis identified key components
based on application domains, such as m
components reveal two major groups of DT
Decision Support (LTDS) DTs. Contrary to ce
AI/ML, real-time capabilities, and bi-directi
built environment. We derived two definit
Both definitions have a must-have compone
components such as prediction, AI, bi-direct

One of the key findings is that the defi
highlighting the need for ongoing revision
address this, we introduce a novel, reprod
the current definitions in response to techn

Data Science (DS) vs Graph Data Science (GDS)





Edsger W. Dijkstra
co-authored 1 paper with
John R. Rice
co-authored 9 papers with
Dan C. Marinescu
co-authored 2 papers with
Theo Lynn
co-authored 7 papers with
Ivanovitch Silva
distance = 4

https://www.csauthors.net/distance/edsger-w-dijkstra/ivanovitch-silva

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We gratefully acknowledge support from the Simons Foundation, member institutions, and all contributors. Donate

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Computer Science > Data Structures and Algorithms

[Submitted on 23 Apr 2025 (v1), last revised 30 Jul 2025 (this version, v2)]

Breaking the Sorting Barrier for Directed Single-Source Shortest Paths

Ran Duan, Jiayi Mao, Xiao Mao, Xinkai Shu, Longhui Yin

We give a deterministic $O(m \log^{2/3} n)$-time algorithm for single-source shortest paths (SSSP) on directed graphs with real non-negative edge weights in the comparison-addition model. This is the first result to break the $O(m + n \log n)$ time bound of Dijkstra's algorithm on sparse graphs, showing that Dijkstra's algorithm is not optimal for SSSP.

Comments: 17 pages
Subjects: **Data Structures and Algorithms (cs.DS)**
ACM classes: F.2.2
Cite as: arXiv:2504.17033 **[cs.DS]**
(or arXiv:2504.17033v2 **[cs.DS]** for this version)
https://doi.org/10.48550/arXiv.2504.17033

Submission history

From: Ran Duan [view email]
[v1] Wed, 23 Apr 2025 18:26:39 UTC (35 KB)
[v2] Wed, 30 Jul 2025 11:02:48 UTC (49 KB)

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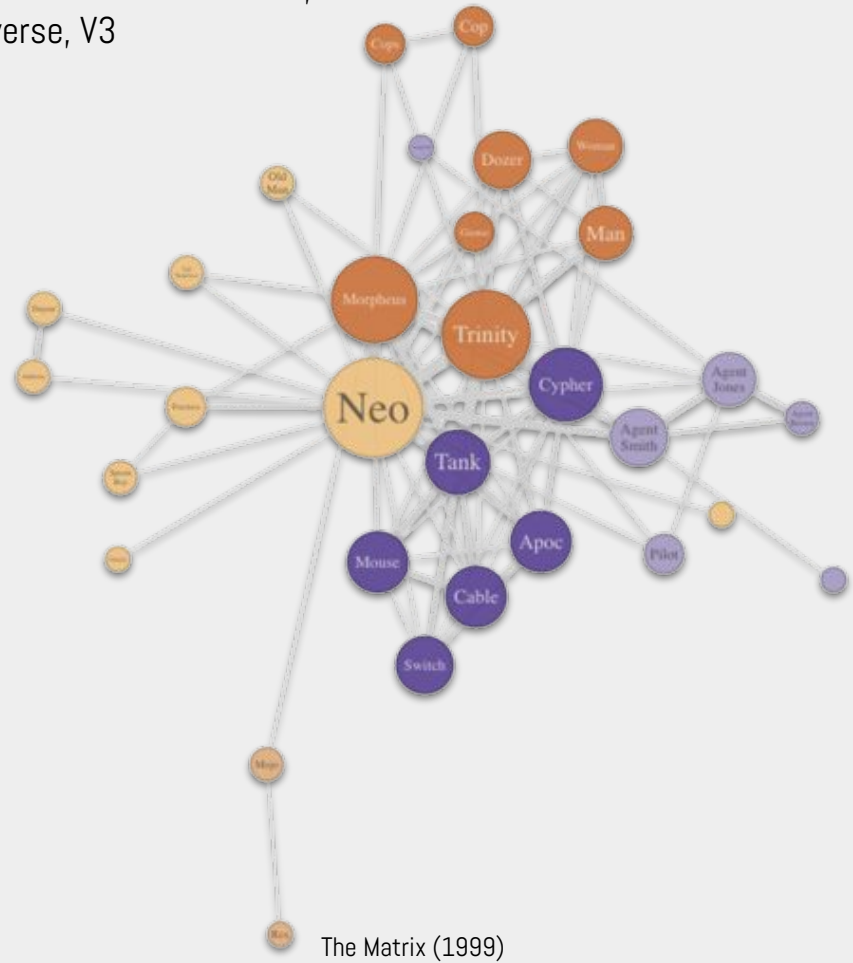
Bike



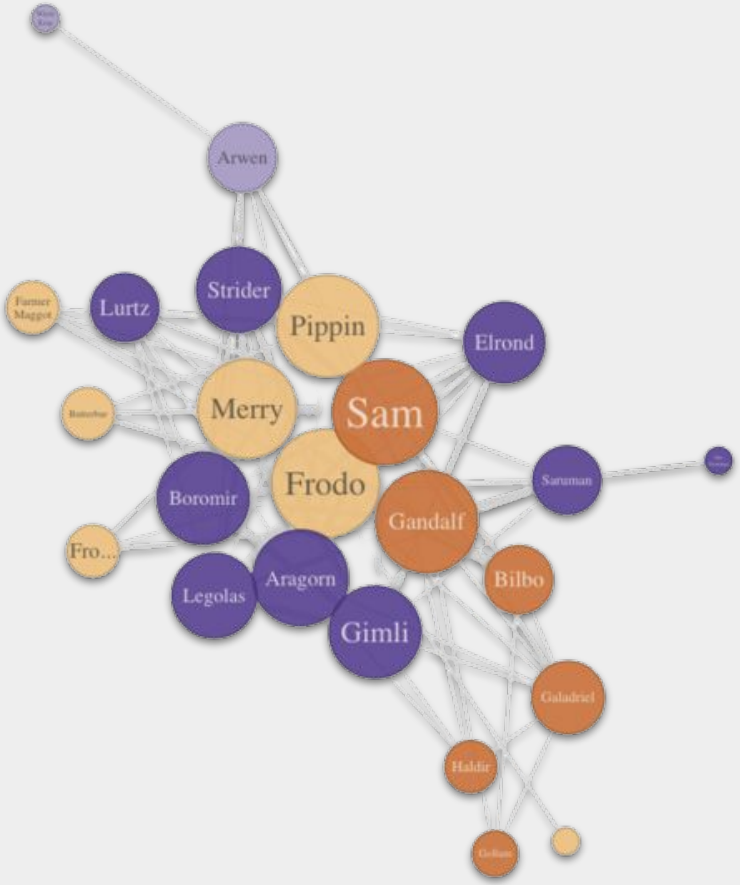
Bike - Drive

Kaminski et al., 2018, "Moviegalaxies - Social Networks in Movies", Harvard Dataverse, V3

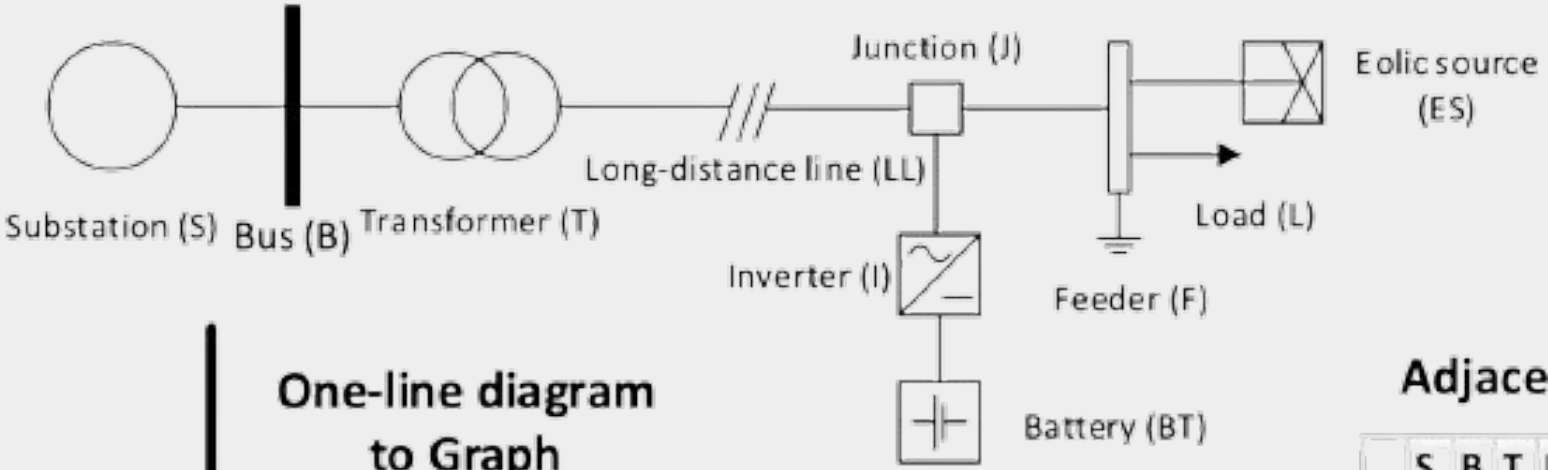
https://moviegalaxies.com



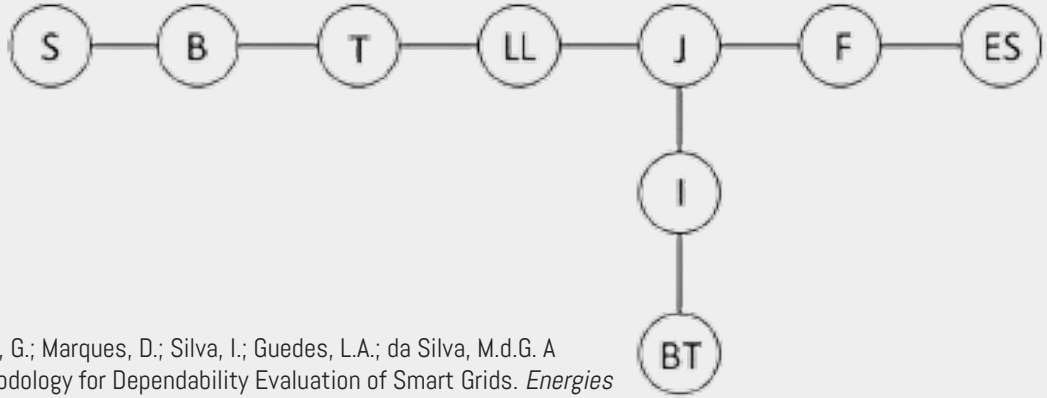
The Matrix (1999)



The Lord of the Rings: The Fellowship of the Ring (2001)



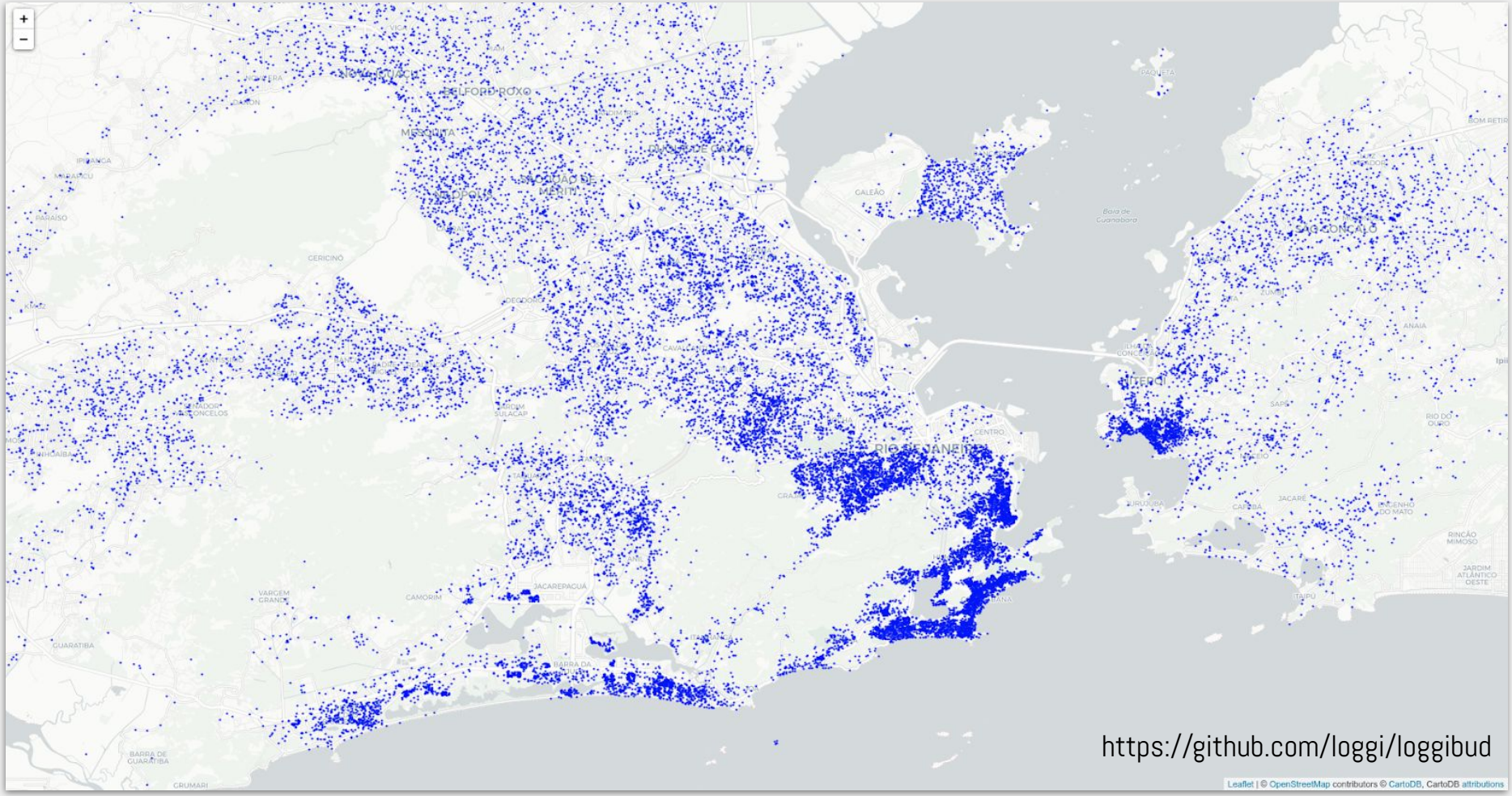
One-line diagram to Graph

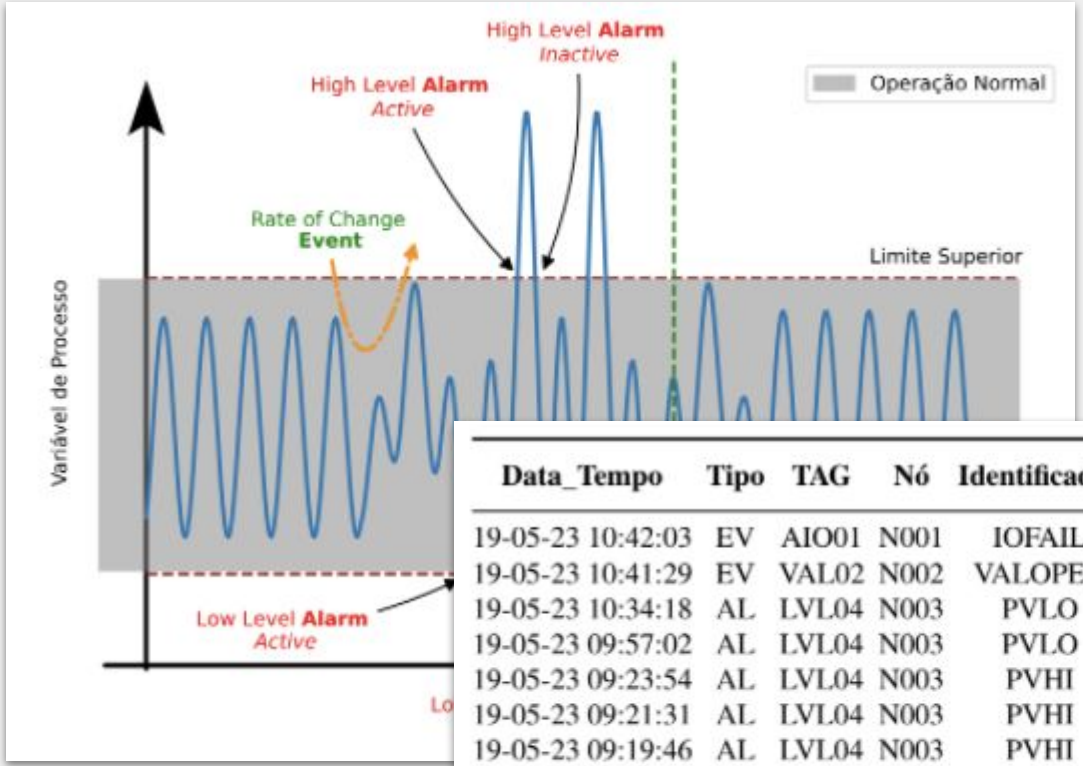


Adjacency Matrix

	S	B	T	LL	J	I	BT	F	ES
S	0	1	0	0	0	0	0	0	0
B	1	0	1	0	0	0	0	0	0
T	0	1	0	1	0	0	0	0	0
LL	0	0	1	0	1	0	0	0	0
J	0	0	0	1	0	1	0	1	0
I	0	0	0	0	1	0	1	0	0
BT	0	0	0	0	0	1	0	0	0
F	0	0	0	0	1	0	0	0	1
ES	0	0	0	0	0	0	0	1	0

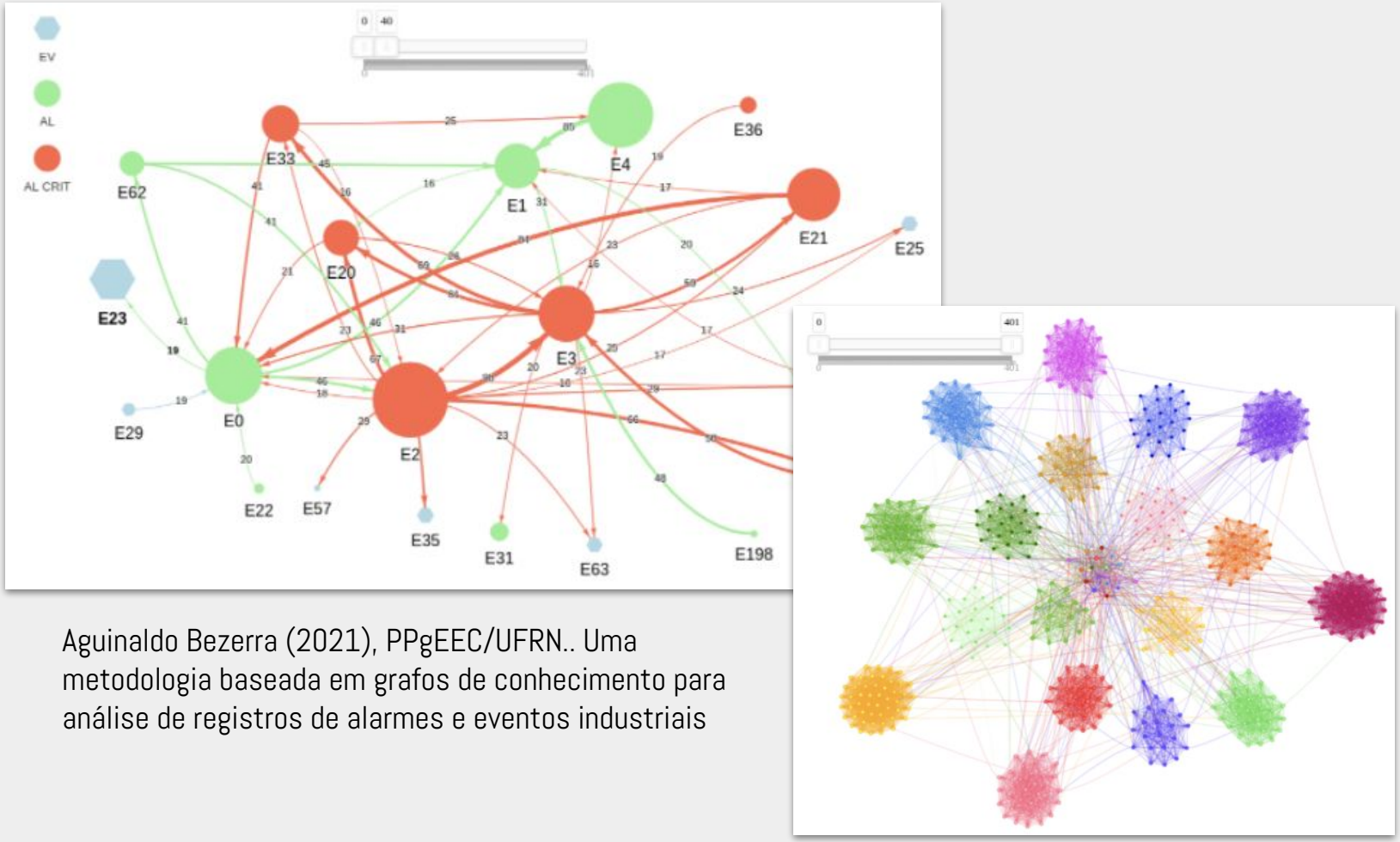
Alves, G.; Marques, D.; Silva, I.; Guedes, L.A.; da Silva, M.d.G. A Methodology for Dependability Evaluation of Smart Grids. *Energies* **2019**, *12*, 1817. https://doi.org/10.3390/en12091817


https://github.com/loggi/loggibud
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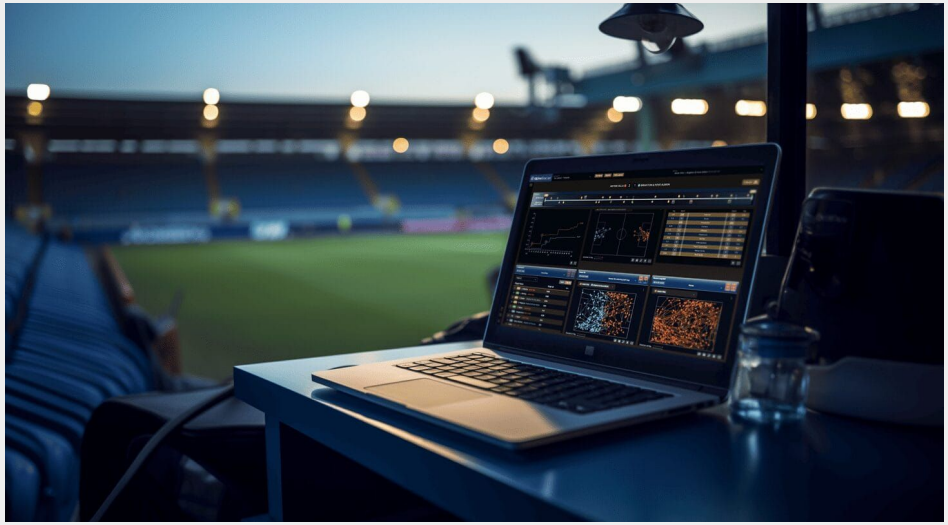


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19-05-23 10:41:29	EV	VAL02	N002	VALOPEN	ACTIVE	INFO	...	Valve Open ...	1045
19-05-23 10:34:18	AL	LVL04	N003	PVLO	INACTIVE	CRITICAL	...	Low Level ...	1037
19-05-23 09:57:02	AL	LVL04	N003	PVLO	ACTIVE	CRITICAL	...	Low Level ...	1031
19-05-23 09:23:54	AL	LVL04	N003	PVHI	INACTIVE	CRITICAL	...	High Level ...	1027
19-05-23 09:21:31	AL	LVL04	N003	PVHI	ACTIVE	CRITICAL	...	High Level ...	1025
19-05-23 09:19:46	AL	LVL04	N003	PVHI	INACTIVE	CRITICAL	...	High Level ...	1022
19-05-23 09:17:34	AL	LVL04	N003	PVHI	ACTIVE	CRITICAL	...	High Level ...	1019
19-05-23 09:15:16	AL	LVL04	N003	PVLO	INACTIVE	CRITICAL	...	Low Level ...	1017
19-05-23 09:12:57	AL	LVL04	N003	PVLO	ACTIVE	CRITICAL	...	Low Level ...	1010
19-05-23 09:04:22	EV	LVL04	N003	PVRATE	ATIVO	WARNING	...	Rate of Chan...	0996

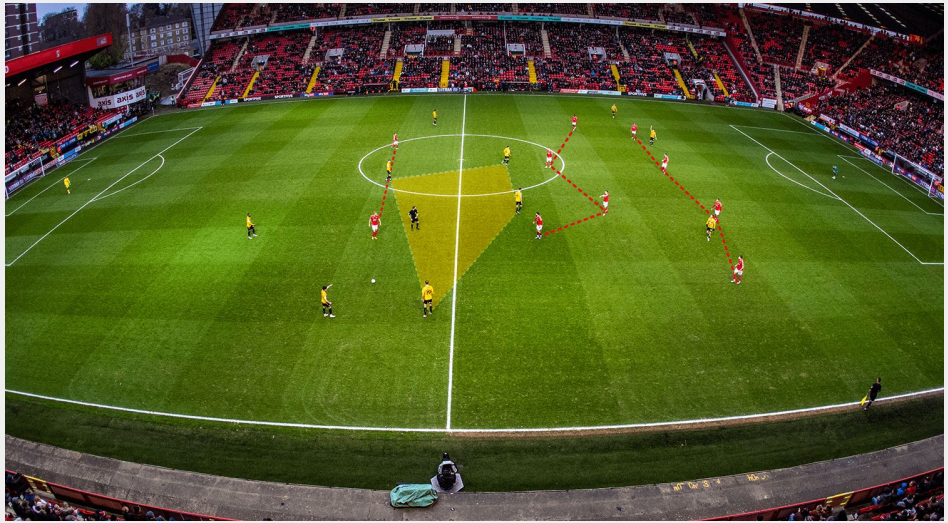
Aguinaldo Bezerra (2021), PPgEEC/UFRN. Uma metodologia baseada em grafos de conhecimento para análise de registros de alarmes e eventos industriais

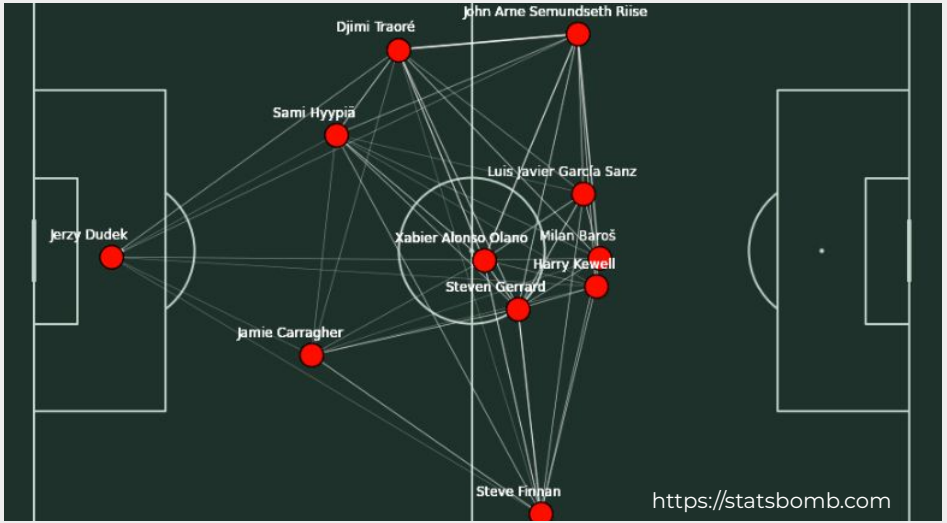


Aguinaldo Bezerra (2021), PPgEEC/UFRN.. Uma metodologia baseada em grafos de conhecimento para análise de registros de alarmes e eventos industriais




360
360




John Arne Semundseth Riise
Djimi Traoré
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Milán Baroš
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Home > Tecnologia

Brasileiro cria ferramenta que cruza dados públicos para mapear risco financeiro de políticos

Sistema usa CPF de agentes públicos e já sinalizou como risco potencial dezenas de milhões de reais ao conectar emendas, licitações e empresas ligadas a familiares

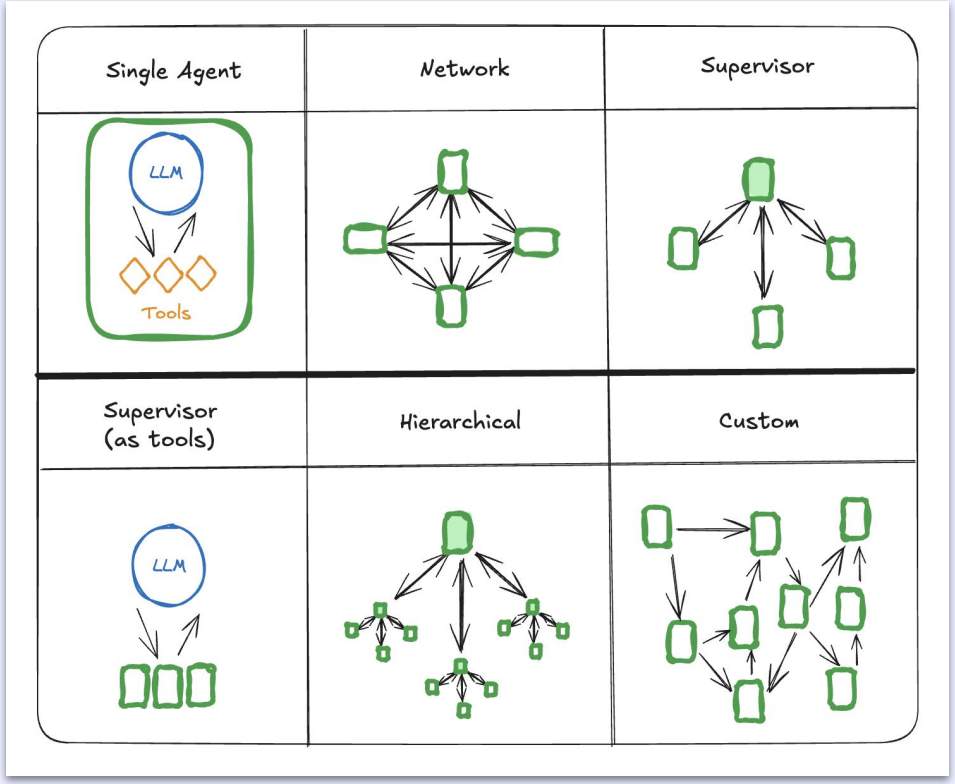






https://github.com/World-Open-Graph/br-acc

Multi-Agents Architecture



Andrej Karpathy
@karpathy

LLM Knowledge Bases

Something I'm finding very useful recently: using LLMs to build personal knowledge bases for various topics of research interest. In this way, a large fraction of my recent token throughput is going less into manipulating code, and more into manipulating knowledge (stored as markdown and images). The latest LLMs are quite good at it. So:

Data ingest:
I index source documents (articles, papers, repos, datasets, images, etc.) into a raw/ directory, then I use an LLM to incrementally "compile" a wiki, which is just a collection of .md files in a directory structure. The wiki includes summaries of all the data in raw/, backlinks, and then it categorizes data into concepts, writes articles for them, and links them all. To convert web articles into .md files I like to use the Obsidian Web Clipper extension, and then I also use a hotkey to download all the related images to local so that my LLM can easily reference them.

IDE:
I use Obsidian as the IDE "frontend" where I can view the raw data, the the compiled wiki, and the derived visualizations. Important to note that the LLM writes and maintains all of the data of the wiki, I rarely touch it directly. I've played with a few Obsidian plugins to render and view data in other ways (e.g. Marp for slides).

Q&A:
Where things get interesting is that once your wiki is big enough (e.g. mine on some recent research is ~100 articles and ~400K words), you can ask your LLM agent all kinds of complex questions against the wiki, and it will go off, research the answers, etc. I thought I had to reach for fancy RAG, but the LLM has been pretty good about auto-maintaining index files and brief summaries of all the documents and it reads all the important related data fairly easily at this ~small scale.

Output:
Instead of getting answers in text/terminal, I like to have it render markdown files for me, or slide shows (Marp format), or matplotlib images, all of which I then view again in Obsidian. You can imagine many other visual output formats depending on the query. Often, I end up "filing" the outputs back into the wiki to enhance it for further queries. So my own explorations and queries always "add up" in the knowledge base.

Linting:
I've run some LLM "health checks" over the wiki to e.g. find inconsistent data, impute missing data (with web searchers), find interesting connections for new article candidates, etc., to incrementally clean up the wiki and enhance its overall data integrity. The LLMs are quite good at suggesting further questions to ask and look into.

Extra tools:
I find myself developing additional tools to process the data, e.g. I vibe coded a small and naive search engine over the wiki, which I both use directly (in a web ui), but more often I want to hand it off to an LLM via CLI as a tool for larger queries.

Further explorations:
As the repo grows, the natural desire is to also think about synthetic data generation + finetuning to have your LLM "know" the data in its weights instead of just context windows.

TLDR: raw data from a given number of sources is collected, then compiled by an LLM into a .md wiki, then operated on by various CLIs by the LLM to do Q&A and to incrementally enhance the wiki, and all of it viewable in Obsidian. You rarely ever write or edit the wiki manually, it's the domain of the LLM. I think there is room here for an incredible new product instead of a hacky collection of scripts.

5:42 PM · Apr 2, 2026 · **21.7M** Views

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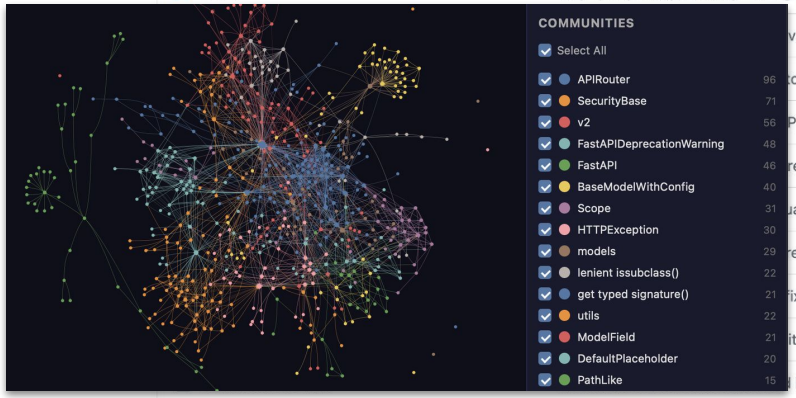
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v8 12 Branches 188 Tags Go to file Code

safishamsi and claude chore: bump to 0.9.40 26128ab · 19 hours ago 1,376 Commits

.github	ci: add PyPI trusted-publishing workflow (publish.yml)	last month
docs	docs: point all website links at graphify.com	last month
graphify	fix(export): make graph.html title marker-truncation take p...	19 hours ago
scripts	docs(readme): add animated "path lights up" demo to See...	last month
tests	fix(export): use portable path in graph.html document title	19 hours ago
	variant runnable on Powe...	4 days ago
	tore files	2 months ago
	P transport to MCP serve...	2 months ago
	refer to #2253	2 weeks ago
	uage extractor split (#1212)	2 months ago
	re split for all platforms (g...	2 months ago
	fix AGENTS.md rebuild co...	4 months ago
	ith Mermaid architecture ...	3 months ago
	l link it from the README	last month
CHANGELOG.md	chore: bump to 0.9.40	19 hours ago
Dockerfile	feat(serve): add Streamable HTTP transport to MCP serve...	2 months ago



About

Turn any codebase, with its docs, SQL schemas, configs, and PDFs, into a queryable knowledge graph. A /graphify skill for Claude Code, Cursor, Codex, and Gemini CLI: local deterministic AST parsing, every edge explained, no vector store.

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Readme

Apache-2.0, MIT licenses found

Security policy

Activity

Custom properties

105.5k stars

355 watching

10.3k forks

Report repository



Agosto - 2026

D	S	T	Q	Q	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Setembro - 2026

D	S	T	Q	Q	S	S
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27	28	29	30			

Outubro - 2026

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Novembro - 2026

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Dezembro - 2026

D	S	T	Q	Q	S	S
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6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Unit 01

Week 01
10/08 - Re-Enroll/Planing
12/08 - Course Outline

Week 02
17/08 - Network Elements I
19/08 - Network Elements II

Week 03
24/08 - Small Worlds I (BFS, DFS, Wiki)
26/08 - Small Worlds II (Path, Dist., Clust. GCC, WCC)

Week 04
31/08 - Hubs I (metrics)
02/09 - Hubs II (core decomposition)

Week 05
07/09 - No class
09/09 - Asynchronous (data viz)

Week 06
14/09 - Applications (use cases)
16/09 - Project Description

Weeks 07, 08, 09, 10, 11 (Hands on Project)

21/09 - Project
23/09 - Project

28/09 - Project
30/09 - Project

05/10 - Project
07/10 - Project

12/10 - No class
14/10 - Review/Consulting

19/10 - Project Presentation (paper, slides, repo)
21/10 - Project Presentation (paper, slides, repo)

End of Unit 01 (21/10)



Unit 02

Week 12
26/10 - Classical algorithms (Dijkstra)
28/10 - No class (videos)

Week 13
02/11 - No class (videos)
04/11 - Classical algorithms (Dijkstra + Heap)

Week 14
09/11 - Classical algorithms (A star)
11/11 - Classical algorithms (Kruskal)

Week 15
16/11 - Review and Questions
18/11 - Exam (multiple choice)

End of Unit 02 (18/11)



Unit 03

Week 16
23/11 - Exam Correction, Grades and Final Project Description
25/11 - Project

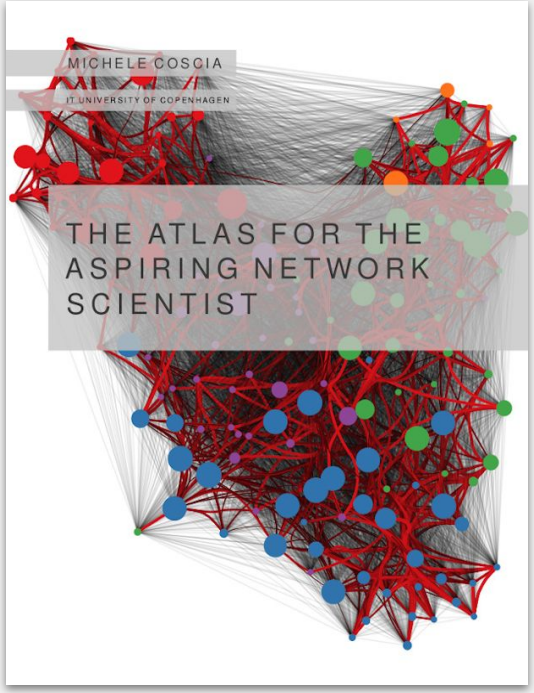
Week 17
30/11 - Project
02/12 - Project Monitoring (checking)

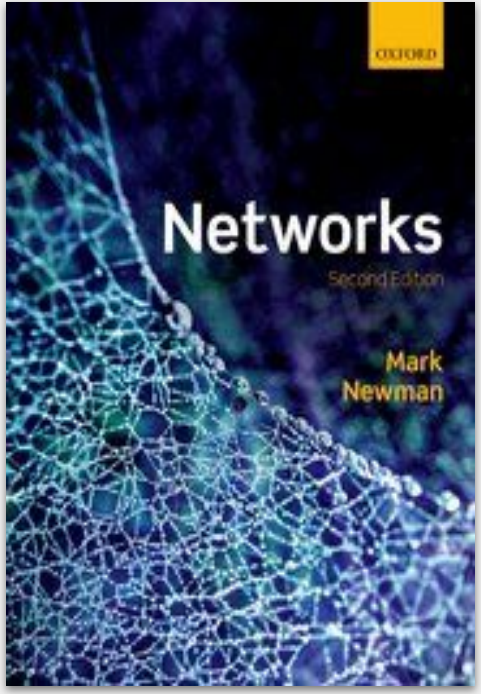
Week 18
07/12 - Project Monitoring (checking)
09/12 - Project Submission

End of Unit 03 (09/12)

Final Exam (16/12)

References


MICHELE COSCIA
IT UNIVERSITY OF COPENHAGEN
THE ATLAS FOR THE ASPIRING NETWORK SCIENTIST


OXFORD
Networks
Second Edition
Mark Newman

MINISTÉRIO DA EDUCAÇÃO
UNIVERSIDADE FEDERAL DO RIO GRANDE DO NORTE

RESOLUÇÃO Nº 016/2023-CONSEPE, de 04 de julho de 2023.

Atualiza o Regulamento dos Cursos de Graduação da Universidade Federal do Rio Grande do Norte - UFRN.

O VICE-REITOR DA UNIVERSIDADE FEDERAL DO RIO GRANDE DO NORTE faz saber que o Conselho de Ensino, Pesquisa e Extensão, no uso das atribuições que lhe confere o art. 17, Inciso III, do Estatuto da UFRN,

CONSIDERANDO o art. 207 da Constituição Federal ao determinar que as universidades gozam de autonomia didático-científica, administrativa e de gestão financeira e patrimonial, e obedecerão ao princípio de indissociabilidade entre ensino, pesquisa e extensão;

CONSIDERANDO a Lei nº 9.394/1996, que estabelece as diretrizes e bases da educação nacional;

CONSIDERANDO a necessidade de atualizar as normas relativas ao ensino de graduação, conforme determina o art. 359 da Resolução no 171/2013 - CONSEPE, de 5 de novembro de 2013, que aprovou o Regulamento dos Cursos Regulares de Graduação; e

CONSIDERANDO o que consta no Processo nº 23077.066985/2023-42,

RESOLVE:

Art. 1º Atualizar o Regulamento dos Cursos de Graduação da Universidade Federal do Rio Grande do Norte – UFRN.

TÍTULO I
DAS DISPOSIÇÕES PRELIMINARES

Art. 2º Este Regulamento tem por finalidade normatizar o ensino de graduação da Universidade Federal do Rio Grande do Norte - UFRN.

Parágrafo único. Os cursos que não possuem oferta regular serão regidos, conforme suas

Art. 110. Não existe abono de faltas, devendo haver compensação do conteúdo e reposição das avaliações realizadas para casos específicos previstos na legislação.

Art. 120. O critério de assiduidade em uma disciplina presencial é satisfeito quando o estudante cumpre a frequência mínima correspondente a 75% (setenta e cinco por cento) da carga horária do componente curricular, considerando o rendimento acadêmico exigido.

Clone me!!!!

https://github.com/ivanovitchm/datastructure